

Nikhil Prabhu

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Professional Summary

Eager to combine an interdisciplinary foundation with expertise in materials science, mechanics and thermodynamics, advanced characterization, data processing and analysis, multiscale and multiphysics materials modeling, and high performance scientific software development to build physics simulation tools and analyze product concepts and designs, while continuing to learn and collaborate across disciplines.

Overview of Skills

- **Interpersonal:** Public speaking, networking, cross-disciplinary teamwork & collaboration, mentoring.
- **Computational Materials Science:** Multiscale and multiphysics modeling, Finite elements and spectral methods.
- **High Performance Computing:** Slurm, GNU Parallel, shell scripting, HDF5, Hybrid MPI/OpenMP Parallelism.
- **Data Processing & Analysis:** Python libraries, statistical analysis, large-scale data processing.
- **Software Engineering:** Python & Fortran development, C++ (course work), Linux, DAMASK Multiphysics and ImageD11 contributions; Git; CI/CD; CMake; unit testing; documentation and debugging.
- **Scientific Toolkits:** ANSYS, COMSOL Multiphysics, LS-DYNA, LS-OPT, CATIA V5, MTEX, ParaView.

Education

Doctor of Engineering Science in Materials Engineering

Jan 2021 – Oct 2025

Departments of Computer Science & Materials Engineering, KU Leuven, Belgium

- Thesis: **Micromechanical Modeling Strategies Toward High-Fidelity Digital Twins of Polycrystals**
Enhanced mesoscale modeling approaches for integration into multiscale material digital twins, enabling accurate predictions for efficient design and development cycles. DOI: 10.5281/zenodo.17465716
- Gained fundamental understanding of crystalline material behavior across time and length scales, from dislocation mechanisms to macroscopic property evolution, strengthening expertise in **multiscale modeling**.
- Translated engineering problems into computational models by designing simulations, applying simplifications for tractability, and validating predictions against physical insights.
- Gained proficiency in **spectral methods** and **finite element analysis** for solving nonlinear ODEs, **crystal plasticity** modeling, and setting efficient computational workflows for large-scale simulations.
- Routinely used high-performance computing clusters, working with the **Slurm** job scheduler, **GNU parallel**, **Shell** scripting, virtual environments, **HDF5** data pipelines, and post-processing in **Python & ParaView**.
- Contributed to the development of the **DAMASK multiphysics** simulation toolkit by translating mathematical constitutive descriptions into modern **Fortran** modules, implementing and debugging solver routines and the **Python** pre- and post-processing tools.
- Gained experience with **software engineering** practices in a scientific codebase, including collaborative version control with **Git**, **CI** pipelines, **CMake**-based build configurations, unit testing, and structured documentation.
- Built strong skills in Python **data processing & analysis** through routine use of **NumPy**, **SciPy**, **Pandas**, **Matplotlib**, and **scikit-learn** for statistical analysis and insight extraction from large datasets.
- Assisted in teaching a *project course in mathematical engineering* and a course on *hot forming techniques*, guiding students and supporting course activities.
- Participated in multiple workshops organized by **Vlaams Supercomputer Centrum** on topics including **scientific software development**, **high-performance computing**, and **data analysis**.

Master of Science in Mechanical Engineering

Sep 2018 – Sep 2020

Linköping University, Sweden (Grade: 4.6 / 5, Pass with Distinction)

- Thesis: **Material Parameter Identification of a Thermoplastic using Full-Field Calibration**
Developed a routine for elastoplastic and damage parameters identification using image-based full-field experimental data at **IKEA Components** (Älmhult, SE). **DiVA id:** diva2:1471392
- Built a strong foundation in the computational description of complex material behavior through coursework in **continuum mechanics**, **heat transfer**, and **numerical methods** (FEM, FDM).
- Applied computational techniques to practical engineering problems through projects involving multiphysics and material damage simulations using **ANSYS**, **COMSOL Multiphysics**, **LS-DYNA**, and **CATIA V5**.
- Completed a group project from **Siemens Energy** (Finspång, SE) involving simulation-driven design of an auxiliary liquid-fuel distribution system operating in a high-temperature multiphysics environment.

Bachelor of Engineering in Mechanical Engineering

Aug 2014 – June 2018

NMAM Institute of Technology, Karnataka, India (GPA: 8.49 / 10, Distinction)

- Thesis: **Design and Fabrication of tilting steering mechanism for Adapted Automobiles**
Implemented a tiltingsteering mechanism on a two-wheeler vehicle to enable maneuvering through controlled body-weight shifting.

Experience

Postdoctoral Researcher

Nov 2025 – Present

Department of Computer Science, KU Leuven, Belgium

- Project: **Advanced modelling and characterization for power semiconductor materials and technologies**
Multiscale and multiphysics modeling of damage nucleation in thick **Cu metallizations** of **SiC** and **GaN** power semiconductors during power cycles.
- Processed large-scale raw data from **scanning 3DXRD**, a state-of-the-art diffraction technique, using HPC clusters (at ESRF, Grenoble) and parallelization.
- Developing thermo-mechanical degradation simulations of Cu interconnects to validate against experiments.
- Contributing to open-source scientific software: **DAMASK Multiphysics** and **ImageD11**.

Master Thesis Student

Jan 2020 – Sep 2020

IKEA Components AB, Älmhult, Sweden

- Simulated thermoplastic material behavior using **LS-DYNA** and performed parameter identification with **LS-OPT**, supported by Python automation scripts.
- Completed specialized software training at **DYNAmore Nordic** (now part of **ANSYS** in Linköping, SE) to strengthen simulation and optimization expertise.

Teaching and Lab Assistant (part time)

Sep 2019 – Jan 2020

Department of Management & Engineering, Linköping University, Sweden

- Assisted in the *Introduction to Computational Mechanics* masters course, supporting labs and exercise sessions; received recommendation from the instructor (Prof. Carl-Johan Thore).

Awards and Contributions

- Attended and presented research contributions at several conferences across Europe and North America, creating opportunities for scientific exchange and collaboration.
- Received a 50% tuition fee merit scholarship from Linköping University for excellence in academic performance during the masters programme.
- Awarded a merit scholarship during the bachelors programme, which included interpersonal skills and communication development workshops organized by the World Konkani Center, India.